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Schottky device and method of manufacturing the same

Abstract

A Schottky device includes a silicon carbide (SiC) substrate of a first conductivity type, a drift layer of the first conductivity type, a trench, a barrier layer of a second conductivity type, an electrically conductive material that at least partially fills the trench and contacts the barrier layer, a first electrode, and a second electrode. The drift layer is formed of SiC and is situated onto the SiC substrate. The trench extends from the top surface of the drift layer towards the SiC substrate. The barrier layer contacts the drifting layer and covers a sidewall and a bottom wall of the trench. The first electrode forms a Schottky junction with the drift layer and forms a low resistivity contact with the barrier layer and the electrically conductive material. The second electrode forms an ohmic contact with the SiC substrate.

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Background/Summary

CLAIM TO DOMESTIC PRIORITY (1) The present application is a continuation of U.S. patent application Ser. No. 16/740,224, now U.S. Pat. No. 11,217,707, filed Jan. 10, 2020, which is a continuation of U.S. patent application Ser. No. 15/705,242, filed Sep. 14, 2017, which applications are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to a semiconductor device, and more particularly to a Schottky device.

BACKGROUND OF THE INVENTION

(2) Schottky devices include a Schottky junction formed between a semiconductor layer and a metal layer. The Schottky devices are characterized by a lower forward voltage drop and a faster switching action compared with general PN junction diodes.

(3) Silicon carbide (SiC) is a wide-gap semiconductor material with a forbidden band gap between

2.35 eV and 3.28 eV. A SiC-based semiconductor is thermally, chemically and mechanically stable and is suitable for applications where high power, high speed, high temperature is required.

(4) New devices and processes that assist in advancing technological needs and industrial applications in semiconductor field are desirable.

SUMMARY OF THE INVENTION

(5) One example embodiment provides a Schottky device. The Schottky device includes a silicon carbide (SiC) substrate of a first conductivity type, a drift layer, a trench, a barrier layer of a second conductivity type opposite the first conductivity type, an electrically conductive material, a first electrode, and a second electrode. The drift layer is formed of SiC and is situated onto the SiC substrate. The drift layer has the first conductivity type with an impurity concentration lower than that of the SiC substrate. The drift layer has a top surface and a bottom surface with the bottom surface contacting the SiC substrate. The trench extends from the top surface of the drift layer towards the SiC substrate. The barrier layer contacts the drifting layer and covers a sidewall and a bottom wall of the trench. The electrically conductive material at least partially fills the trench and contacts the barrier layer. The first electrode is situated on the top surface of the drift layer. The first electrode forms a Schottky junction with the drift layer and forms a low resistivity contact with the barrier layer and the electrically conductive material. The second electrode forms an ohmic contact with the SiC substrate. In an example embodiment, the barrier layer includes a buffer junction barrier (JB) sub layer with a first impurity concentration and a JB layer with a second impurity concentration. The first impurity concentration is lower than the second impurity concentration.

(6) Other example embodiments are discussed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a Schottky device with a planar structure in accordance with an example embodiment;
- (2) FIG. 1B illustrates a junction barrier (JB) region enclosed by more than one buffer JB sub region in accordance with an example embodiment;
- (3) FIG. 2 illustrates a top view of JB regions and buffer JB sub regions in accordance with an example embodiment;
- (4) FIG. 3 illustrates a top view of JB regions and buffer JB sub regions in accordance with another example embodiment;
- (5) FIG. 4A shows a graph illustrating a performance comparison between a Schottky device with a planar structure in accordance with an example embodiment and a conventional Schottky device;
- (6) FIG. 4B is a table showing values for current leakage for a Schottky device with a planar structure in accordance with an example embodiment and a conventional Schottky device;
- (7) FIG. 5 illustrates a Schottky device with a shallow trench in accordance with an example embodiment;
- (8) FIG. 6 illustrates a Schottky device with a deep trench in accordance with an example embodiment;
- (9) FIG. 7 illustrates a Schottky device with a deep trench in accordance with another example embodiment;
- (10) FIG. 8 illustrates a method that manufactures a Schottky device in accordance with an example embodiment;
- (11) FIG. 9A illustrates forming a JB region by using an oxide layer as a hard mask in accordance with an example embodiment;
- (12) FIG. 9B illustrates forming a buffer JB sub region by using an oxide layer as a hard mask in

accordance with an example embodiment;

(13) FIG. 10 illustrates a method that manufactures a Schottky device in accordance with an example embodiment; and

(14) FIG. 11 illustrates a method that manufactures a Schottky device in accordance with an example embodiment.

DETAILED DESCRIPTION OF THE DRAWINGS

(15) Example embodiments relate to Schottky devices with novel structure designs and improved performances, such as improved leakage and breakdown voltages.

(16) Conventional or existing Schottky devices or diodes are disadvantageous in their high current leakage and low breakdown voltage under reverse bias. Conventional Schottky devices thus face limitations in many high power applications.

(17) Furthermore, limitations exist regarding application of Silicon (Si)-based manufacturing processes to Silicon carbide (SiC)-based devices due to differences in their material properties. For example, for a Si-based process, an initial doping profile is generally formed with ion implantation. Initial overlap of the initial doping profile is not smooth and needs a thermal anneal to facilitate dopant redistribution to form an expected profile. However, this does not work well for SiC material as there is almost no dopant redistribution in SiC material during thermal anneal. The doping profile remains choppy and weak spots are formed that increase current leakage.

(18) One or more example embodiments solve one or more of the technical problems associated with conventional Schottky devices as set forth above. One or more example embodiments achieve Schottky devices with improved device performance by providing newly-designed device structures and manufacturing processes to contribute to semiconductor technology.

(19) One or more example embodiments achieve Schottky devices with improved performance, such as reduced leakage and improved breakdown voltage. For example, one example embodiment demonstrates a Schottky device with leakage approximately five times lower than that of its conventional counterparts. Advantages of example embodiments further include the proposed novel processes for manufacturing Schottky devices, which can be implemented with one or more benefits of simplicity, flexibility, and better alignment, etc. These novel processes overcome the non-transferability of Si-based process when manufacturing SiC-based devices. Schottky devices of example embodiments have broader and better applications compared with conventional Schottky devices.

(20) Example embodiments includes a Schottky device with a planar structure, a shallow trench structure, and a deep trench structure respectively. A junction barrier (JB) region is enclosed by a buffer JB sub region with a lower impurity concentration compared with the JB region. Both the JB region and the buffer JB sub region have the conductivity type opposite that of a drift region. The buffer JB sub region is designed or configured such that a regulated doping junction is formed between the JB region and the Schottky region. As such, the maximal electric field between the JB region and Schottky region as well as the metal Schottky barrier is reduced, thereby improving overall performance for the Schottky device.

(21) One or more example embodiments recognize that leakage mostly happens at the bottom and corners of the JB region and the surface of the Schottky junction or area under the Schottky metal. Enclosure of the JB region by the buffer JB sub region at the bottom (i.e., bottom enclosure) is designed to be larger than that at the sidewall of the JB region or lateral enclosure. The lateral enclosure and the bottom enclosure can be independently adjusted or obtained, for example, by designing process parameters. For example, the lateral enclosure can be designed optimally to reduce electric field at the sidewall of the JB region on one hand, and maintain a low turn-on resistance for the Schottky junction on the other hand, thereby achieving optimal performance. Furthermore, by having a tighter buffer JB sub region at the sidewall, forward voltage drop for the Schottky junction is prevented from increasing and can be retained at a low value, which contributes to fast switching speed and lower power consumption.

(22) One or more example embodiments include novel manufacturing process for the Schottky devices. The JB regions and the buffer JB sub regions are formed by ion implantations without relying on or attributable to dopant redistribution. For Schottky devices with a planar structure or a shallow trench structure, a single oxide hard mask can be employed in formation of both the JB regions and the buffer JB sub regions with ion implantations, thereby self-alignment is achieved. For Schottky devices with a deep trench structure, a novel process including epitaxial growth is provided with simplicity and flexibility.

(23) FIG. 1A illustrates a Schottky device **100** with a planar structure in accordance with an example embodiment.

(24) The Schottky device **100** includes a substrate **110**, a drift layer or region **120**, one or more junction barrier (JB) regions or layers **126**, one or more buffer JB sub regions or layers **128**, an anode or first electrode or first metal layer **130**, and a cathode or second electrode or second metal layer **140**.

(25) By way of example, the substrate **110** has a first surface **112** and a second surface **114** opposite the first surface **112**. The substrate **110** includes or is formed of silicon carbide (SiC) of a first conductivity type (e.g., n-type). As an example, the substrate **110** is highly n-doped SiC with an impurity or doping concentration of $5.0 \times 10^{19} \text{ cm}^{-3}$ (i.e., $5.0 \times 10^{19} \text{ cm}^{-3}$) or greater.

(26) By way of example, the drift layer **120** is situated or disposed on the substrate **110**. The drift layer **120** has a top surface **122** and a bottom surface **124**. The bottom surface **124** contacts the first surface **112** of the substrate **110**.

(27) The drift layer **120** includes dopants or impurities of the first conductivity type with an impurity concentration lower than that of the substrate **110**. As an example, the drift layer **120** is n-doped with a concentration in a range from $5.0 \times 10^{14} \text{ cm}^{-3}$ to $2.0 \times 10^{16} \text{ cm}^{-3}$ and has a thickness range from 5 micrometer (μm) to 80 μm . In an example embodiment, the impurity concentration of the drift layer **120** is $8.5 \times 10^{15} \text{ cm}^{-3}$. As another example, the drift layer **120** is n-doped SiC that is epitaxially grown onto the substrate **110**.

(28) By way of example, the JB region **126** is included in the drift layer **120** and extends from the top surface **122** towards the substrate **110**. The JB region **126** has a width $w_{\text{sub.J}}$ and a depth $d_{\text{sub.J}}$ beneath the top surface **122**. In an example embodiment, the width $w_{\text{sub.J}}$ varies in a range from 0.5 μm to 4.0 μm , and the depth $d_{\text{sub.J}}$ varies in a range from 0.3 μm to 1.0 μm .

(29) The JB region **126** includes dopants of a second conductivity type (e.g., p-type) opposite the first conductivity type and has a first impurity concentration. For example, the JB region **126** can be formed by ion implantation.

(30) By way of example, the buffer JB sub region **128** is included in the drift layer **120** and extends from the top surface **122** towards the substrate **110**. The buffer JB sub region **128** has a width $w_{\text{sub.b}}$ and a depth $d_{\text{sub.b}}$. In an example embodiment, the depth $d_{\text{sub.b}}$ is in a range from 0.7 μm to 1.5 μm . The width $w_{\text{sub.b}}$ is larger than the width $w_{\text{sub.J}}$ by 0.6 μm to 2.0 μm (i.e., lateral enclosure of the JB region **126** by the buffer JB sub region **128** is in a range from 0.3 μm to 1.0 μm per side).

(31) The buffer JB sub region **128** encloses the JB region **126** with a lateral enclosure dimension (i.e., $(w_{\text{sub.b}} - w_{\text{sub.J}})/2$) and a bottom enclosure dimension (i.e., $d_{\text{sub.b}} - d_{\text{sub.J}}$). The lateral enclosure dimension is smaller than the bottom enclosure dimension. The lateral enclosure dimension and the bottom enclosure dimension are adjustable independent on each other.

(32) By way of example, if the lateral enclosure dimension is too small, the buffer JB sub region cannot effectively shield the Schottky junction under reverse bias and cannot effectively reduce the maximal electric field at the interface of the JB region and the Schottky region (As used herein, the "Schottky region" refers to the region between the JB regions in the drift layer **120** where Schottky junction is formed at the surface), thereby being not effective in reducing leakage and increasing breakdown voltage. On the other hand, if the lateral enclosure dimension is too large, it will reduce the area of Schottky junction and thus reduce the forward current of the Schottky device. The

capability of independently adjusting the lateral enclosure dimension in accordance with example embodiments are favorable or useful because an optimal lateral enclosure dimension can be achieved without negatively influencing other device parameters, such as depths and impurity concentrations of the JB regions and the buffer JB sub regions. Schottky device in accordance with example embodiments can be designed and manufactured with flexibility.

(33) By way of example, as illustrated in FIG. 1A, the buffer JB sub region **128** fully encloses the JB region **126** in relation to sidewall and bottom wall. In this manner, the buffer JB sub region **128** provides effective shielding for the JB region **126** such that the maximal electric field between the JB region **126** and the Schottky region is reduced and overall performance for the Schottky device is improved.

(34) By way of example, the buffer JB sub region **128** is of the second conductivity type with a second impurity concentration that is smaller than the first impurity concentration. In an example embodiment, the first impurity concentration is in a range from $7.0E18 \text{ cm}^{-3}$ to $6.0E20 \text{ cm}^{-3}$, and the second impurity concentration is in a range from $1.0E17 \text{ cm}^{-3}$ to $6.0E18 \text{ cm}^{-3}$.

(35) In one example embodiment, there are more than one set of JB regions and buffer JB sub regions. The distance $w_{sub.0}$ between two adjacent buffer JB sub regions **128** from two adjacent sets of JB regions and buffer JB sub regions respectively is in a range from 2 μm to 6 μm . As an example, $w_{sub.0}=3 \text{ }\mu\text{m}$.

(36) By way of example, the JB region **126** and the buffer JB sub region **128** can include dopants of same or different species. In one example embodiment, the JB region **126** includes dopants of aluminum (Al), and the buffer JB sub region **128** includes one or more dopants selected from a group consisting of Al, Boron (B), and Gallium (Ga).

(37) By way of example, the anode **130** is situated on the top surface **122** of the drift layer **120**. The anode **130** forms a Schottky junction with exposed portions of the top surface **122** of the drift layer **120**, and forms a low resistivity contact with the JB region **126**. The anode **130** can be a metal layer, such as aluminum (Al), titanium (Ti), nickel (Ni), or can be a multi-metal structure formed of two or more metals. In an example embodiment, the anode **130** includes multiple metal structures that are short-circuited.

(38) By way of example, the cathode **140** contacts the second surface **114** to form a low resistivity contact with the substrate **110**. As an example, the cathode **140** includes one or more metals, such as nickel (Ni), titanium (Ti), silver (Ag), platinum (Pt), gold (Au) or the like that is capable of forming an ohmic contact with the SiC substrate.

(39) Although FIG. 1A illustrates three sets of JB region and buffer JB sub region, and each set illustratively includes one JB region and one buffer JB sub region, a person having ordinary skill in the art would appreciate that each set can include multiple layers of buffer JB sub regions with regulated doping profiles according to practical requirements. This is illustrated in FIG. 1B. FIG. 1B illustrates a JB region **126**, and three buffer JB sub regions **1281**, **1282**, and **1283** in the drift layer **120** in accordance with an example embodiment. The impurity concentration decreases from the JB region **126** towards the buffer JB sub region **1283**.

(40) By way of example, the shapes of the JB region and the buffer JB sub region from a top or plane view can be predetermined or designed. For example, the shapes can be circular, square, hexagonal, strip, or other suitable shapes according to practical requirements. This is illustrated in FIGS. 2 and 3.

(41) FIG. 2 illustrates a top view of JB regions and buffer JB sub regions in accordance with an example embodiment. As illustrated, both the JB regions **226** and the buffer JB sub regions **228** are square shape on the top surface of the drift layer **220**.

(42) FIG. 3 illustrates a top view of JB regions and buffer JB sub regions in accordance with another example embodiment. As illustrated, both the JB regions **326** and the buffer JB sub regions **328** are circular shape on the top surface of the drift layer **320**.

(43) The Schottky device with structure as shown in FIG. 1A reduces maximum electric field at the junction formed at an interface between the JB region and the drift layer or Schottky region, thereby reducing reverse current leakage and increasing breakdown voltage.

(44) The shapes as shown in FIGS. 2 and 3 are for illustrative purpose only. A person having ordinary skill in the art would appreciate that the shapes for the JB regions and the buffer JB sub regions from a top view can be other suitable shapes, such as rectangular shape, stripe shape etc., according to practical requirements.

(45) FIG. 4A shows a graph 410 illustrating a performance comparison between a Schottky device with structure with reference to FIG. 1A (“Buffer JBS”) and a conventional Schottky device (“Std JBS”). FIG. 4B is a table 420 showing values of leakage current.

(46) The “Buffer JBS” includes a SiC drift layer of 11um in thickness and being n-doped with an impurity concentration of $8.5 \times 10^{15} \text{ cm}^{-3}$. $d_{\text{sub.J}}=0.4 \text{ um}$, $w_{\text{sub.J}}=2.0 \text{ um}$, $d_{\text{sub.b}}=0.9 \text{ um}$, $w_{\text{sub.b}}=3.0 \text{ um}$, $w_{\text{sub.0}}=3.0 \text{ um}$. The breakdown voltage is above 1200V.

(47) The “Std JBS” includes only a JB region without a Buffer JB sub region. The width and depth for the JB region of the “Std JBS” are 3.0 um and 0.9 um respectively. The JB region of the “Std JBS” has a same impurity concentration as that for the JB region of the “Buffer JBS”.

(48) As shown in FIGS. 4A and 4B, compared with the median leakage current at 1200V reverse bias for “Std JBS”, the median leakage current for the “Buffer JBS” is 2.6 uA, which is about 5 times lower than that for the “Std JBS” (12.2 uA). In addition, the leakage currents for “Buffer JBS” group are more stable with less variation compare to the standard JBS structure.

(49) FIG. 5 illustrates a Schottky device 500 with a shallow trench in accordance with an example embodiment.

(50) For purpose of clarify, details that are same as or similar to the description with reference to FIG. 1A are omitted or stated briefly hereinafter. Elaboration is mainly focused on differences.

(51) As illustrated, the Schottky device 500 includes a substrate 510 with a first surface 512 and a second surface 514, a drift layer 520 with a top surface 522 and a bottom surface 524, one or more trenches 525, one or more junction barrier (JB) regions 526, one or more buffer JB sub regions 528, an anode or first electrode or first metal layer 530, and a cathode or second electrode or second metal layer 540. By way of example, both the substrate 510 and the drift layer 520 include SiC material of a first conductivity type (e.g., n-type).

(52) By way of example, the trench 525 is formed in the drift layer 520 and extends from the top surface 522 of the drift layer 520 towards the substrate 510. The trench 525 has a width $w_{\text{sub.st}}$ and a depth $d_{\text{sub.st}}$ beneath the top surface 522 of the drift layer 520. In an example embodiment, the width $w_{\text{sub.st}}$ is in a range from 0.5 um to 3.0 um, and the depth $d_{\text{sub.st}}$ is in a range from 0.5 um to 1.5 um. The trench 525 is at least partially filled with one or more electrically conductive materials 529, such as p-doped poly-Si, metal (e.g., Al, Ti, Ni, etc), and the like.

(53) By way of example, the JB region 526 is included in the drift layer 520 and extends from the top surface 522 towards the substrate 510. The JB region 526 encloses the trench 525 in relation to the sidewall and bottom of the trench 525. The JB region 526 has a width $w_{\text{sub.J}}$ and a depth $d_{\text{sub.stJ}}$ beneath the bottom of the trench 525. As an example, the depth $d_{\text{sub.stJ}}$ is in a range from 0.8 um to 1.8 um. The width $w_{\text{sub.J}}$ is in a range from 1 um to 4 um. In an example embodiment, the JB region 526 encloses the trench 525 with a lateral enclosure in a range from 0.25 um to 0.5 um.

(54) By way of example, the JB region 526 includes dopants of a second conductivity type (e.g., p-type) opposite the first conductivity type and with a first impurity concentration.

(55) By way of example, the buffer JB sub region 528 is included in the drift layer 520 and extends from the top surface 522 towards the substrate 510. The buffer JB sub region 528 has a width $w_{\text{sub.b}}$ and a depth $d_{\text{sub.stb}}$ beneath the bottom of the trench 525. As an example, the depth $d_{\text{sub.stb}}$ is in a range from 1.1 um to 2.4 um. The width $w_{\text{sub.b}}$ is in a range from 1.5 um to 5 um. In an example embodiment, the buffer JB sub region 528 encloses the JB region 526 with a lateral

enclosure in a range from 0.25 μm to 0.5 μm .

(56) The buffer JB sub region **528** encloses the JB region **526** with a lateral enclosure dimension (i.e., $(w.\text{sub.b}-w.\text{sub.J})/2$) and a bottom enclosure dimension (i.e., $d.\text{sub.stb}-d.\text{sub.stJ}$). The lateral enclosure dimension is smaller than the bottom enclosure dimension. The lateral enclosure dimension can be adjusted independent on the bottom enclosure dimension.

(57) The buffer JB sub region **528** includes dopants of the second conductivity type (e.g., p-type) and with a second impurity concentration. The doping concentration for the buffer JB sub region **528** is lower than that for the JB region **526** such that maximal electric field at PN junctions between the JB region **526** and the drift layer **520** under reverse voltage is reduced, thereby improving overall device performance, such as leakage and breakdown voltage.

(58) In an example embodiment, the JB region **526** and the buffer JB sub region **528** have same dopant specie selected from a group consisting of Aluminum (Al), Boron (B), and Gallium (Ga). In another example embodiment, the JB region **526** and the buffer JB sub region **528** have different dopant species selected from a group consisting of Al, B, and Ga.

(59) By way of example, there are more than one set of trenches, JB regions and buffer JB sub regions, the distance w_1 between two adjacent buffer JB sub regions from two adjacent sets respectively is in a range from 2 μm to 6 μm .

(60) In an example embodiment, the buffer JB sub region **528** includes plural buffer JB sub regions with each having a different impurity concentration.

(61) By way of example, the first metal layer **530** disposed on the drift layer **520**. The first metal layer **530** forms a Schottky junction with exposed portions of the drift layer **520**, and forms a low resistivity contact with the JB region and the one or more electrically conductive materials **529**.

(62) By way of example, the second metal layer **540** form an ohmic contact with the SiC substrate **510**.

(63) In an example embodiment, thickness of drift layer=12.0 μm , $w.\text{sub.st}=1.0$ μm , $w.\text{sub.b}=2.0$ μm , $d.\text{sub.st}=0.8$ μm , $d.\text{sub.stJ}=1.1$ μm , $d.\text{sub.stb}=1.5$ μm , $w.\text{sub.1}=3.0$ μm , and the SiC Schottky device with such parameters has a breakdown voltage of greater than 1200V.

(64) FIG. 6 illustrates a Schottky device **600** with a deep trench in accordance with an example embodiment.

(65) As illustratively shown, the Schottky device **600** includes a substrate **610**, a drift layer **620**, one or more trenches **625**, one or more barrier layers or regions **627**, an anode or first metal layer **630**, and a cathode or second metal layer **640**.

(66) By way of example, the substrate **610** has a first surface **612** and a second surface **614**. The substrate **610** includes SiC of a first conductivity type (e.g., n-type).

(67) By way of example, the drift layer **620** is situated or disposed on the substrate **610**. The drift layer **620** has a top surface **622** and a bottom surface **624**. The drift layer **620** is of the first conductivity type with an impurity concentration lower than that of the substrate **610**. In an example embodiment, the thickness of the drift layer **620** is equal to or greater than 6.5 μm .

(68) By way of example, the trench **625** is formed in the drift layer **620** and extends from the top surface **622** towards the substrate **610**. The trench **625** has a width $w.\text{sub.dtB}$ and a depth $d.\text{sub.dtB}$ beneath the top surface **622** of the drift layer **620**. As an example, the width $W.\text{sub.dtB}$ is in a range from 1.0 μm to 3.0 μm . The depth $d.\text{sub.dtB}$ is in a range from 1.5 μm to 5.0 μm .

(69) Inside the trench **625** forms (e.g., by selective epitaxial growth) the barrier layer or region **627** of a second conductivity type (e.g., p-type). The barrier layer **627** covers the sidewall and the bottom wall of the trench **625** and contacts the drift layer **620**.

(70) By way of example, the barrier layer **627** has a doping profile with an impurity concentration increasing from the drift layer **620** towards one or more electrically conductive materials **629** that at least partially fills the trench **625**. In an example embodiment, the barrier layer **627** is an epitaxial layer formed of one or more materials selected from a group consisting of Gallium nitride (GaN), and Silicon Carbide (SiC).

(71) In an example embodiment, the barrier layer **627** includes a buffer junction barrier (JB) sub layer with a first impurity concentration and a JB layer with a second impurity concentration. The buffer JB sub layer is sandwiched between the drift layer **620** and the JB layer. The first impurity concentration is lower than the second impurity concentration. This will be detailed later with reference to FIG. 7.

(72) The rest of the trench **625** is at least partially filled with the one or more electrical materials **629**. As used herein, “the rest of the trench” refers to internal space of the trench excluding the space occupied by the barrier layer. The barrier layer **627** is sandwiched between the one or more electrical materials **629** and the sidewall, the bottom wall of the trench **625**. In some example embodiments, the one or more electrical materials **629** are unnecessary. Instead, the whole trench **625** is filled with the barrier layer **627**. In some other embodiments, the barrier layer **627** includes both a JB region and a buffer JB sub region.

(73) In an example embodiment, the rest of the trench **625** has a width $w_{sub.dt}$ and a depth $d_{sub.dt}$ beneath the top surface **622**. As an example, the width $w_{sub.dt}$ is in a range from 0 μm to 2 μm . The depth $d_{sub.dt}$ is in a range from 0 μm to 2.5 μm .

(74) In an example embodiment including more than one trench, the distance $w_{sub.2}$ between two adjacent trenches is in a range from 2 μm to 6 μm .

(75) By way of example, the first metal layer **630** is situated on the drift layer **620**. the first metal layer **630** forms a Schottky junction with exposed portions of the top surface **622** of the drift layer **620**, and forms a low resistivity contact with the barrier layer **627** and the one or more electrically conductive materials **629**.

(76) By way of example, the second metal layer **640** forms an ohmic contact with the SiC substrate **610**.

(77) In an example embodiment, thickness of drift layer=12.5 μm , $w_{sub.dt}$ =0.6 μm , $w_{sub.dtB}$ =1.6 μm , $d_{sub.dt}$ =1.7 μm , $d_{sub.dtB}$ =2.2 μm , $w_{sub.2}$ =3.0 μm , and the SiC Schottky device with such parameters has a breakdown voltage of greater than 1200V.

(78) FIG. 7 illustrates a Schottky device **700** with a deep trench in accordance with another example embodiment.

(79) As illustratively shown, the Schottky device **700** includes a substrate **710** with a first surface **712** and a second surface **714**, a drift layer **720** with a top surface **722** and a bottom surface **724**, one or more trenches **725**, one or more electrically conductive materials **729** filled in the trench **725**, an anode or first metal layer **730**, and a cathode or second metal layer **740**.

(80) Compared with the Schottky device **600** with reference to FIG. 6, differences include that the barrier layer or region of Schottky device **700** includes a JB layer **726** and a buffer JB sub layer **728**. The buffer JB sub layer **728** is sandwiched between the drift layer **720** and the JB layer **726**. The buffer JB sub layer **728** covers the sidewall and the bottom wall of the trench **725**.

(81) As illustrated, the trench **725** has a width $w_{sub.dB}$ and a depth $d_{sub.dB}$ beneath the top surface **722**. The JB layer has a width $w_{sub.dJ}$ and a depth $d_{sub.dJ}$ beneath the top surface **722**. The rest of the trench **725** has a width w_{an} and a depth d_{an} . The distance between two adjacent trenches is $w_{sub.3}$.

(82) In an example embodiment, the buffer JB sub layer **728** has a thickness of 300 nm to 1000 nm (i.e., $(w_{sub.dB}-w_{sub.dJ})/2=300_{sup.}\sim 1000$ nm) and the JB layer **726** has a thickness of 200 nm to 1200 nm (i.e., $(w_{sub.dJ}-w_{sub.dt1})/2=200_{sup.}\sim 1200$ nm).

(83) FIG. 8 illustrates a method **800** that manufactures a Schottky device in accordance with an example embodiment. As an example, the method **800** is used to manufacture or produce the Schottky device **100** with reference to FIG. 1A with improved device performance.

(84) Block **802** states providing a SiC substrate. For example, the SiC substrate is highly-doped with dopants of a first conductivity type, such as n-type dopants (e.g., Nitrogen, phosphorus, arsenic, etc.).

(85) Block **804** states epitaxially growing a SiC drift layer. For example, the SiC drift layer is

formed onto the SiC substrate with epitaxial growth with in-situ doping of the first conductivity type dopants.

(86) Block **806** states forming a junction barrier (JB) region. For example, the JB region is formed by implanting dopants or impurities of a second conductivity type, such as p-type dopants (e.g., boron, aluminum, gallium, etc.), into a predetermined area in the drift layer. The JB region is formed attributable to ion implantations rather than impurity diffusion process. Regulated doping profile is formed by adjusting parameters such as ion beam energy, implantation dose, tilt and twist angles, etc. A single implantation process, or multiple or repeated implantation processes can be employed according to practical requirements.

(87) By way of example, the JB region is formed by using an oxide hard mask. For example, a thick oxide layer (e.g., silicon oxide) is grown or deposited onto the top surface of the drift layer. Photoresist layer is formed (e.g., by spin-coating) onto the oxide layer and patterned with a mask such that portions of the oxide layer is isotropic etched to form one or more openings for selective implantation. Through these openings, by using the patterned oxide layer as a hard mask, dopant ions are implanted into the drift layer to form the JB region.

(88) Usage of the patterned oxide layer as a hard mask is illustrated in FIG. **9A**. As illustrated, the oxide layer is patterned (e.g., by wet etching or dry etching, or combination of both) to expose portions of the top surface of the drift layer **920** disposed on the substrate **910** such that ions or dopants **904A** are implanted into the drift layer **920** to form the JB region **926**. The remaining portions of the top surface are covered by oxide **902A** to prevent ions from reaching the drift layer **920**.

(89) In one example embodiment, a chain of implantations is conducted to form an expected doping profile. In another example embodiments, one or more doping species are doped to form the JB region.

(90) Block **808** states forming a buffer JB sub region. For example, the buffer JB sub region is formed by implanting dopants or impurities of the second conductivity type, such as p-type. The buffer JB sub region is formed attributable to ion implantations rather than impurity diffusion process. Regulated doping profile is formed by adjusting parameters such as ion beam energy, implantation dose, tilt angles, etc. A single implantation process, or a chain of implantation processes can be employed according to practical requirements.

(91) By way of example, the buffer JB sub region is formed by using the same oxide layer as hard mask to achieve self-alignment. In an example embodiment, one or more wet chemical etching or isotropic etching processes are conducted on the oxide hard mask to trim or widen the openings to further open 0.3 μm to 0.5 μm per side, followed by an ion implantation process. Trimming the openings of the oxide hard mask is illustrated in FIG. **9B**. The openings are widened by a predetermined size such that less oxide **902B** is left coving the top surface of the drift layer **920** and more portions are exposed for ion implantation. Then ions **904B** are implanted into the drift layer **920** to form the buffer JB sub region **928**.

(92) In an example embodiment, multiple trimming steps of the oxide hard mask are conducted. To obtain a tailored doping profile, the openings are trimmed or tailored for multiple times from small to large by wet etching or isotropic etching. Each time trimming of the openings is followed by an ion implantation process. In some example embodiments, process parameters such as ion beam energy, implantation dose, tilt and twist angle, dopant species or the like are adjusted each time such that expected doping depth, concentration and the like are achieved. In one example embodiment, with patterned oxide layer as a hard mask, p-type dopants (e.g., boron, aluminum, etc.) are implanted into the drift layer at a concentration of about $1.0\text{E}17\text{ cm}^{-3}$ to $6.0\text{E}18\text{ cm}^{-3}$.

(93) In one example embodiment, instead of using an oxide hard mask, plural separate masks are employed to trim the openings such that the openings are sized from small to large for selective implantation to form the JB region and the buffer JB sub region.

(94) Although ion implantation is shown in FIGS. 9A and 9B has a direction perpendicular to the top surface of the drift layer, a person having ordinary skill in the art would appreciate that implantation angle can be tilted according to practical applications.

(95) In one example embodiment, the dopants in the JB region and the buffer JB sub region are thermally annealed at a temperature of 1600 degrees Celsius ($^{\circ}$ C.) or above. As an example, dopants are annealed or activated in a furnace with a carbon cap layer. A sacrifice oxidation at dry oxygen (O.sub.2) at 900 $^{\circ}$ C. to 1000 $^{\circ}$ C. for 15 min to 30 min is conducted to remove carbon and silicon segregation formed at the SiC surface in the JB region after the high temperature anneal.

(96) As another example, dopants are activated with microwave rapid thermal annealing (RTA) at 2000 $^{\circ}$ C. to 2100 $^{\circ}$ C. for a time period in a range from 5 seconds to 60 seconds. As a further example, dopant activation is conducted by direct pulse laser annealing (PLA) with wavelength at 308 nm to 355 nm at 1.5 J/cm.sup.2 to 3.4 J/cm.sup.2 either at room temperature or at evaluated temperature up to 800 $^{\circ}$ C. and with either single shot or multiple anneal shots.

(97) Block **810** states forming an anode metal layer. The anode metal layer can be a single metal layer or multiple metal structures.

(98) In an example embodiment, after stripping off the oxide hard mask or oxide layer, a metal layer is deposited onto the drift region. The metal layer serves as the Schottky metal at the drift region, and server as the metal contact at the JB region. A rapid thermal annealing (RTA) is conducted at a medium temperature at 500 $^{\circ}$ C. to 650 $^{\circ}$ C. for 2 min to enhance the Schottky contact formation ideal factor.

(99) Block **812** states forming a cathode metal layer. For example, the cathode metal layer is formed on a side of the substrate opposite the drift layer. The cathode metal layer includes a metal, such as Ni, Ti, Ag, Pt, or Au that is capable of forming a low resistivity contact with n-type silicon carbide.

(100) FIG. **10** illustrates a method **1000** that manufactures a Schottky device in accordance with an example embodiment. For example, the method **1000** can be used to manufacture the Schottky device **500** with reference to FIG. **5**.

(101) Block **1002** states providing a SiC substrate of a first conductivity type. For example, the SiC substrate is highly doped with dopants of a first conductivity type, such as n-type dopants (e.g., nitrogen, phosphorus, arsenic, etc.).

(102) Block **1004** states epitaxially growing a drift layer of the first conductivity type on the SiC substrate. For example, the SiC drift layer is formed onto the SiC substrate with epitaxial growth with in-situ doping of the first conductivity type dopants.

(103) Block **1006** states forming a trench with dry etching. As an example, an oxide layer (e.g., silicon oxide) can be used as hard mask for dry etching (e.g., reactive-ion etching) to form the trench. For purpose of trench formation, the oxide layer can be patterned with one or more openings to form the hard mask. As another example, other separate mask can be used for dry etching to form the trench.

(104) Block **1008** states forming a junction barrier (JB) region of a second conductivity type with ion implantation. As an example, the JB region is formed by trimming or tailoring or widening openings of the same oxide layer to expose more portions of the surface of the drift layer and then conducting one or more ion implantation steps. As another example, the step of trimming followed by ion implantation can be repeated for multiple times according to practical requirements.

(105) Block **1010** states forming a buffer JB sub region of a second conductivity type with ion implantation. For example, many of statements with reference to block **808** of FIG. **8** are applicable to block **1010**.

(106) For example, an oxide layer is deposited onto the drift layer as a hard mask. Then the oxide layer is etched to form one or more openings. The one or more openings are trimmed by isotropic etching the oxide layer such that dopant ions of the second conductivity type are selectively implanted into the drift layer to form the JB region. Subsequently, the one or more opening are

further trimmed by isotropic etching the oxide layer such that the dopant ions of the second conductivity type are selectively implanted into the drift layer to form the buffer JB sub region.

(107) Block **1012** states filling the trench. For example, the trench is filled with one or more electrically conductive materials, such as p-doped poly-Si, metal (e.g., Al, Ti, Ni, etc), and the like.

(108) Block **1014** states forming a first metal layer on a top surface of the drift layer. For example, many of statements with reference to block **810** of FIG. **8** are applicable to block **1014**.

(109) Block **1016** states forming a second metal layer on a second surface of the SiC substrate. For example, the second surface is a surface of the SiC substrate opposite the drift layer. For example, many of statements with reference to block **812** of FIG. **8** are applicable to block **1016**.

(110) FIG. **11** illustrates a method **1100** that manufactures a Schottky device in accordance with an example embodiment. For example, the method **1100** can be used to manufacture the Schottky device **600** with reference to FIG. **6** or the Schottky device **700** with reference to FIG. **7**.

(111) Block **1102** states providing a SiC substrate of a first conductivity type. For example, the SiC substrate is highly doped with dopants of a first conductivity type, such as n-type dopants (e.g., phosphorus, arsenic, etc.).

(112) Block **1104** states forming, on the SiC substrate and by epitaxial growth, a drift layer including SiC of the first conductivity type. For example, the SiC drift layer is formed onto the SiC substrate with epitaxial growth with in-situ doping of the first conductivity type dopants.

(113) Block **1106** states forming a trench. As an example, an oxide layer (e.g., silicon oxide) can be used as hard mask for dry etching (e.g., reactive-ion etching) to form the trench. For purpose of trench formation, the oxide layer can be patterned with one or more openings to form the hard mask. As another example, other separate mask can be used for dry etching to form the trench.

(114) Block **1108** states epitaxially growing a barrier layer of a second conductivity type. For example, in-situ epitaxial growth with p-type doping is conducted to grow the barrier layer or region.

(115) In one example embodiment, multiple epitaxial layers are grown with each layer having different impurity concentration. As an example, the multiple epitaxial layers include a JB layer and one or more buffer JB sub layers.

(116) In another example embodiment, to form the barrier region, a first doped layer or buffer JB sub layer is formed by selective epitaxial growth (e.g., Metal-organic Chemical Vapor Deposition (MOCVD)) of doped SiC, or GaN, on sidewall and bottom wall of the trench. A second doped layer or JB layer is formed by selective epitaxial growth of doped SiC or GaN onto the first doped layer. The second doped layer has a higher impurity concentration than that of the first doped layer.

(117) Block **1110** states filling the trench. For example, the trench is filled with one or more electrically conductive materials, such as p-doped poly-Si, metal (e.g., Al, Ti, Ni, etc.), and the like. In an example embodiment, the trench is filled with a Low-Pressure Chemical Vapor Deposition (LPCVD), Metal Organic Chemical Vapor Deposition (MOCVD), Atomic Layer Deposition (ALD) or the like. An etch back or chemical mechanical polished (CMP) process is conducted to bring the final trench fill plug 50 nm to 150 nm above the top surface of the drift layer.

(118) Block **1112** states forming an anode with a first metal layer on a top surface of the drift layer. For example, many of statements with reference to block **810** of FIG. **8** are applicable to block **1112**.

(119) Block **1114** states forming a cathode with a second metal layer. For example, many of statements with reference to block **812** of FIG. **8** are applicable to block **1114**.

(120) As used herein, a “low resistivity contact” refers to a contact that readily allows the movement of electric charge. For example, an ohmic contact is one kind of low resistivity contact.

(121) The present invention is illustrated by example embodiments in a non-limiting way. A person having ordinary skill in the art would appreciate description with reference to one example embodiment may be applicable to one or more other example embodiments. A person having

ordinary skill in the art would conceive of one or more variations in accordance with example embodiments as illustrated herein, which still falls into scope of these example embodiments

Claims

1. A semiconductor device, comprising: a substrate; a first semiconductor layer formed over the substrate; a trench having a width and a depth formed into a surface of the first semiconductor layer and removing a portion of the first semiconductor layer defined by the trench; a second semiconductor layer formed within the trench and extending from the surface of the first semiconductor layer to a bottom of the trench; a third semiconductor layer formed over the second semiconductor layer in the trench and extending to the surface of the first semiconductor layer; and an electrically conductive material deposited within the trench and extending to a portion of the third semiconductor layer formed at the bottom of the trench.
2. The semiconductor device of claim 1, further including: a first conductive layer formed over the first semiconductor layer; and a second conductive layer formed over the substrate.
3. The semiconductor device of claim 1, wherein an impurity concentration of the third semiconductor layer is greater than an impurity concentration of the second semiconductor layer.
4. The semiconductor device of claim 1, wherein the substrate includes a silicon carbide layer.
5. The semiconductor device of claim 1, wherein a first portion of the trench is wider than a second portion of the trench.
6. A semiconductor device including a plurality of cells, comprising: a substrate; a first semiconductor layer formed over the substrate; and a first cell formed within the first semiconductor layer, the first cell including, (a) a second semiconductor layer formed within a first trench having a width and a depth in the first semiconductor layer and extending from a surface of the first semiconductor layer to a bottom of the trench, (b) a third semiconductor layer formed over the second semiconductor layer in the first trench and extending to the surface of the first semiconductor layer, and (c) an electrically conductive material deposited within the first trench and extending to the third semiconductor layer.
7. The semiconductor device of claim 6, further including: a first conductive layer formed over the first semiconductor layer; and a second conductive layer formed over the substrate.
8. The semiconductor device of claim 6, wherein an impurity concentration of the third semiconductor layer is greater than an impurity concentration of the second semiconductor layer.
9. The semiconductor device of claim 6, wherein the substrate includes a silicon carbide layer.
10. The semiconductor device of claim 6, wherein a first portion of the first trench is wider than a second portion of the first trench.
11. The semiconductor device of claim 6, further including a second cell formed within the first semiconductor layer, the second cell including, (a) the second semiconductor layer formed within a second trench in the first semiconductor layer, and (b) the electrically conductive material deposited within the second trench over the second semiconductor layer.
12. A method of making a semiconductor device, comprising: providing a substrate; forming a first semiconductor layer over the substrate; forming a trench having a width and a depth into a surface of the first semiconductor layer to remove a portion of the first semiconductor layer defined by the trench; forming a second semiconductor layer within the trench; forming a third semiconductor layer over the second semiconductor layer in the trench and extending to the surface of the first semiconductor layer; and depositing an electrically conductive material within the trench and extending to the third semiconductor layer.
13. The method of claim 12, further including: forming a first conductive layer over the first semiconductor layer; and forming a second conductive layer over the substrate.
14. The method of claim 12, wherein an impurity concentration of the third semiconductor layer is greater than an impurity concentration of the second semiconductor layer.

15. The method of claim 12, wherein the substrate includes a silicon carbide layer.
 16. The method of claim 12, wherein a first portion of the trench is wider than a second portion of the trench.
 17. The method of claim 12, further including providing a second cell formed within the first semiconductor layer, the second cell including, (a) the second semiconductor layer formed within a second trench in the first semiconductor layer, and (b) the electrically conductive material deposited within the second trench over the second semiconductor layer.
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